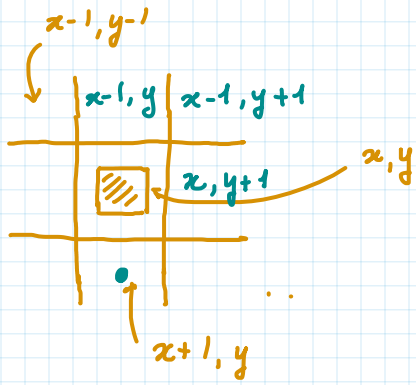
OPERATORI LOCALI DIFFERENZIABILI → I ORDINE : SOBEL
→ II ORDINE : LAPLACIANO

Def: FILTRO di SOBEL è un filtro del I ORDINE di dim $n \in \mathbb{N}$, $n = 3$.
↓
di SHARPENING



$\underline{x}_0 = (x, y)$

$h, k \in \mathbb{Z}$ e $\underline{x} = (x+h, y+k)$

$\underline{x}$ è in un INTORNO $B(\underline{x}_0, r)$

Polinomio di TAYLOR in $f : \mathbb{R}^2 \to \mathbb{C}$:
↓
centrato in (x, y)

$$f(x+h, y+k) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \cdot (\cancel{x} + h - \cancel{x})^1 \cancelto{1}{(y+k-y)^0}$$
$$+ \frac{\partial f}{\partial y}(x,y) \cdot \cancelto{1}{(x+h-x)^0} (\cancel{y} + k - \cancel{y})^1$$
$$\cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \cdot h + \frac{\partial f}{\partial y}(x,y) \cdot k$$

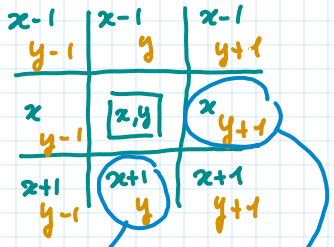
Se $k = 0$:

$$f(x+h, y+k) = f(x+h, y) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \cdot h$$

$$\frac{\partial f}{\partial x}(x,y) \cong \frac{f(x+h,y) - f(x,y)}{h}$$

e $h = 1$:

$$\frac{\partial f}{\partial x}(x,y) \cong f(x+1, y) - f(x,y)$$



Se $h = 0$ e $k = 1$:

$$f(x, y+1) \cong f(x,y) + \frac{\partial f}{\partial y}(x,y) \cdot 1$$

$$\hookrightarrow \frac{\partial f}{\partial y}(x,y) \cong f(x, y+1) - f(x,y)$$